



Large language models (LLMs) as psychotherapists: an analysis based on psychodynamic psychotherapy theory

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Published online: 13 August 2026
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Abstract

The article analyses the capabilities and limitations of large language models' (LLMs) powering chat interfaces, such as ChatGPT, in the context of their potential use as “psychotherapists”. The authors relate this issue to the principles of psychodynamic psychotherapy, in which interpersonal relationships, unconscious processes, transference, and countertransference are of key importance. Based on a review of the literature and technical analysis, it is indicated that contemporary LLMs do not meet the basic conditions for conducting therapy in this field. They lack awareness, emotionality, and the ability to mentalize, whereas their “empathy” is purely linguistic and simulated. Human interactions with the model can create the illusion of a therapeutic relationship, promote anthropomorphizing and excessive trust in the algorithm. The article highlights the ethical, legal, and purely psychological risks associated with assigning chatbots the role of therapists. The analysis encompasses problems of hallucinations, data bias, and lack of ethical oversight. At the same time, the authors point to the potential benefits of using LLMs as support tools – in psychological education, preliminary diagnosis, stress reduction, or as a support for therapists in the process of analysis. They emphasize the need for an interdisciplinary approach combining psychology, law, and ethics, as well as the need to develop standards for safety and oversight of the use of LLMs in the field of mental health. The article concludes with a call for caution in the face of the growing integration of technology into therapeutic practice.

Keywords Psychotherapy · Psychodynamic · Large language models · ChatGPT · Companionship · Therapy · Artificial intelligence

1. Introduction

The rapid advancement of Large Language Models (LLMs) is evident in their widespread applications, powering not only first-party chat interfaces like ChatGPT, Claude, and Gemini, but also proliferating through third-party platforms such as Poe and Character.AI. These advanced artificial intelligence (AI) systems have achieved a level of fluent textual interaction that closely resembles conversation with another human being (Mei et al., 2024; Mittelstädt et al., 2024; Jones & Bergen, 2025). The classic Turing Test, which was designed to examine a machine's ability to mimic humanlike interactions, is becoming less relevant (Gams & Kramar, 2024).

Interactions with advanced AI systems are increasingly perceived as conversations with potential communication partners with personal characteristics, and this impression is reinforced as the conversational quality of models continues to improve (Gams & Kramar, 2024). LLMs match or even

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surpass human capabilities in different tasks (Taloni et al., 2023; Wang, 2023; Webb et al., 2023; Akpan, 2025; Kramer, 2025). This includes more than just a linguistic mimicry, and it has already been pointed out that LLMs can outperform humans in social situational judgments (Mittelstädt et al., 2024). Their widespread use is influencing the dynamics of social communication and the ways in which messages are obtained and processed (Elyoseph et al., 2024). Ways of applying technology are also changing. Parallel to technological progress, the number of users declaring the use of LLMs for mental well-being purposes, including emotional support or seeking therapeutic advice in a direct way, is growing (Zao-Sanders, 2025).

In the 1960s, Joseph Weizenbaum created the ELIZA program, whose task was to simulate conversations between users and a psychotherapist. Although the algorithm was pretty simple and based on turning statements into questions, many participants of the experiments attributed surprising depth of understanding and empathy to ELIZA and anthropomorphize the algorithm (Weizenbaum, 1966; Natale, 2018). The whole experience prompted Weizenbaum “to warn against the seductive power of AI and against the willingness of humans to overestimate the capacity of computers” (Neuberger et al., 2023, p. 2). Almost six decades later, Weizenbaum’s concerns remain relevant, albeit in a much more advanced technological context and in the face of growing challenges for classic psychology.

These shifts require thorough analysis of human-machine interactions. Users already report regular conversations with chatbots of an introspective or emotional nature (Zao-Sanders, 2025). In some cases, models are directly assigned the role of a psychotherapist (Brand & Wegner, 2024). These relationships are characterized by a far-reaching anthropomorphizing (Cohn et al., 2024), excessive faith in LLMs’ judgment (Lee et al., 2024; Zhai et al., 2024), and even addiction to contacts with the chatbots (Lin et al., 2025; Maral et al., 2025). The observed phenomena raise a number of fundamental questions not only about the effectiveness of such forms of mental support, but above all about their ethics, safety, and compatibility with the basic assumptions of psychotherapy as a scientific field and clinical practice.

From the perspective of psychodynamic psychotherapy – an approach that assumes the centrality of interpersonal relationships, the importance of unconscious processes, and the transformative role of transference and countertransference (Gabbard, 2004; McWilliams, 2004) – the question of whether a chatbot can function as a therapist takes on particular significance. However, there is a lack of systematic analyses situating modern LLMs in the light of this particular theoretical school of psychotherapy.

This study provides the necessary theoretical framework for a comprehensive analysis of how LLMs are used

as “psychotherapists”, while raising important questions about the future of psychotherapy as a discipline and the implementation of new technologies as potential tools for increasing the effectiveness of psychotherapists’ work. The reflections rooted in the psychodynamic understanding of the psychotherapeutic process allow for further analysis on ethical or legal grounds.

2. Methodological assumptions

2.1. Research assumptions of the article

The use of LLMs for psychotherapeutic purposes is still a relatively new topic and does not have a firmly established place in psychotherapeutic discourse. This may cause dissonance among psychotherapists who lack clear analyses of the applicability of LLMs in psychotherapeutic processes and guidelines for practical use-cases. Commercial apps are also entering this use-case sphere, but the main approach is grounding them in therapeutic frameworks such as cognitive behavioral therapy. Their clinical validation remains varied.

In this study, we therefore attempt to fill the gap of understanding the extent to which LLMs can (or cannot) act as “therapists”, and to formulate a theoretical framework to understand the limitations and potential dangers associated with such use of technology. This framework can serve as a universal comparative matrix, the applicability of which may increase in the future, especially if AI achieves (some degree of) consciousness (Kurzweil, 1991, 2005; Bostrom, 2022; Bubeck et al., 2023), or due to new understanding of the “self” (Metzinger, 2010).

The article identifies the conditions necessary for the (psychodynamic) psychotherapy process to take place. The capabilities and limitations of LLMs serve as a comparative reference point. We also highlight practical, ethical, and legal risks of using LLMs in the therapeutic process.

Consequently, we focus on three overarching research questions:

RQ1. *To what extent can interaction with a large language model fulfill the conditions of a therapeutic relationship as understood in psychodynamic psychotherapy?*

RQ2. *What are the risks associated with assigning chatbots the role of psychotherapists without adequate grounding in the structure of the psychotherapy process?*

RQ3. *In what areas should LLMs be used to support psychodynamic psychotherapy?*

A critical analysis of LLMs’ capabilities in therapeutic processes is not aimed at demonizing them or fundamentally rejecting their applications. Rather, we point to the necessity to raise social awareness in regard to the differences

between a real psychotherapy and interaction with a chatbot and to subsidiarily apply technology for the therapeutic process. However, the study primarily relates to psychodynamic psychotherapy, which might be seen as a limitation.

2.2. Large language models as psychotherapists – literature review

The use of LLMs in psychotherapy has been the subject of a number of theoretical and empirical studies. Lin et al. (2024) test, for example, the psychological safety of responses and personality dark patterns of LLMs' behaviors. The impact of generative AI on psychology has been analyzed in the works of Demszky et al. (2023), Abdurahman et al. (2024), Ma et al. (2024), Salah et al. (2024), Stade et al. (2024), Heinz et al. (2025), Moore et al. (2025), and Na et al. (2025). Abdurahman et al. (2024) introduce the term "GPTology" to describe quasi-therapy offered by ChatGPT.

Moore et al. (2025) give a negative assessment of the extent to which LLMs can take on the role of psychotherapists, contrasting this with the formal licensing requirements in the US. On the other hand, Lalk et al. (2025) propose improvements in the analysis of therapy session transcripts, and Fung et al. (2024) propose the use of LLMs to imitate patient responses in simulated training sessions.

Hatch et al. (2025) investigate the applicability of LLMs to crisis intervention in couples therapy. Heinz et al. (2025) positively evaluate the effectiveness of a specially designed, fine-tuned Therabot in treating major depressive disorder symptoms, generalized anxiety disorder symptoms, and eating disorders. Foran et al. (2024) compare the potential of LLMs in self-help as an alternative to psychoeducation. There are enough studies of AI-powered conversational agents to drive first meta-analyses. They typically point out small effects and lack of long term/follow-up data (Zhong et al., 2024; Feng et al., 2025).

Some of the research is devoted to the applications of AI, including LLMs, in medicine, which also refers to benchmarking reasoning (Webb et al., 2023; Akpan, 2025; Mei et al., 2024), empathy (Ayers et al., 2023), photo analysis (Kramer, 2025), diagnostic accuracy (Ayers et al., 2023), and social situations judgments (Mittelstädt et al., 2024). Specific applications are recognized in the areas of counteracting the symptoms of schizophrenia (Stefaniak et al., 2019), ADHD, OCD, and substance misuse disorders (Beg & Verma, 2024), counteracting dementia (Treder et al., 2024), and understanding emotional cues for hearing-impaired children (Kushawaha, 2024). De Freitas et al. (2023) and Reis et al. (2024) test patients' trust in AI-based diagnostics. Nori et al. (2025) show that AI can achieve better results in diagnosing complex medical cases due to iterative reasoning.

On the other hand, more attention is being paid to the issue of addiction to LLMs (Ciudad-Fernandez et al., 2025; Maral et al., 2025; Yankouskaya et al., 2025) and over-reliance on their outputs (Lee et al., 2024). An interesting direction of research refers to counteracting the bias exhibited by LLMs using psychological techniques, including a substitute for "psychotherapy for LLMs" proposed by Lin et al. (2024) and Ben-Zion et al. (2025).

2.3. The spread of LLMs' use for psychotherapeutic purposes

Research shows that over 50% of Americans use LLMs (Rainie, 2025). The scale of use is constantly increasing, as are the forms of application. An important trend is the use of so-called AI Companions that play a specific role and act as a companion or partner to the user. Robb and Mann (2025) indicate that 72% of the teenagers surveyed have used the AI Companion function at least once, and 52% declare its regular use. 33% declare that they use them for social interaction and relationships, and 12% directly for mental health support (Robb & Mann, 2025).

The main reasons for using AI Companions include categories such as: "it gives advice", "they're always available when I need someone to talk to", "they don't judge me", "I can say things I wouldn't tell my friends or family", or "it's easier than talking to real people". Each of these descriptions was used by the participants between 9% and 18% of the time (Robb & Mann, 2025).

The report by Internet Matters (2025) confirms these trends. Although only 3% of minors directly indicated seeking therapy by LLMs, the responses "seeking advice", "just chatting", and "I wanted a friend" were indicated by 23%, 18%, and 6% of respondents respectively. Hashem et al. (2025) established that 25% of minor users use chatbots for "chatting and keeping company", whereas 20% search for "getting advice on something personal".

With the widespread use of generative AI tools, the use of LLMs for strictly psychotherapeutic purposes is also growing. Zao-Sanders (2024, 2025) indicates that in 2025, among adults, the most frequently reported use of LLMs is "Therapy/Companionship" category, followed by the "Organize my life" category, which focuses less on experiences and emotions and more on structuring tasks and priorities. Third place in the ranking was "Finding purpose", which again has a profound self-reflection dimension. These meta-analyses reveal significant trends, which cannot be ignored by the practitioners of psychotherapy.

3. Psychotherapy in the psychodynamic approach

3.1. The psychodynamic approach in psychotherapy theory

Psychodynamic psychotherapy is a psychotherapeutic method rooted in psychoanalytic thinking. In this approach, psychopathological syndromes are seen as the result of unconscious conflict, relationship patterns, disadvantageous defense mechanisms, and deficits in affect regulation and reflective functions or capabilities (Gabbard, 2004; Cabaniss et al., 2011). Therefore, the main therapeutic mechanisms include raising awareness of unconscious meanings, interpreting transference, and ‘working through’ established patterns in a live relationship with the therapist (McWilliams, 2004). This school of psychotherapy integrates perspectives of object relations, attachment, and mentalization, emphasizing how internalized models of ‘self-other’ relationships shape current experience.

Psychodynamic work is based on a structured setting (regular meetings, relationship boundaries, confidentiality), on therapist-patient cooperation, and on the therapist’s disciplined use of neutrality and empathy (Gabbard, 2004; McWilliams, 2004). Furthermore, the therapy is based on activation of transference in the therapeutic relationship, the therapist’s recognition and reflective use of countertransference, and the joint development of meanings in interpretation and processing (Gabbard, 2004).

3.2. Necessary conditions

Psychodynamic approach assumes the existence of a set of relational and structural preconditions, so the therapy can occur and be credible and effective. This set was identified in the course of literature review, although it should be noted that specific sets of rules are not common and universal.

Working alliance.

A solid therapeutic alliance is one of the strongest predictors of outcomes in various modalities (Saketopoulou, 1999; Cabaniss et al., 2011). In psychodynamic therapy, the alliance provides a framework that allows for the exploration of transference.

Patient readiness.

Patients need sufficient motivation to cope with discomfort, reality testing, or to work together on shared meanings. Patients should present ego strength to tolerate affects without decompensation (Cabaniss et al., 2011).

Capacity to mentalize/reflect.

Patients must at least be able – often with the help of a therapist – to think about thoughts and feelings (their own and those of others). This reflective function enables the

symbolization of experience and is the goal of therapy in cases of fragility (Brockmann et al., 2025).

Containment.

In the context of psychodynamic psychotherapy theory, one of the key aspects of the therapist’s work is the ability to contain – that is, to consciously accept, process, and return to the patient their difficult, often unconscious emotional states in a way that is understandable and leads to change (Gabbard, 2004; Cabaniss et al., 2011). The foundation of this process is the existence of a subject capable of mentalization, i.e., consciously experiencing, understanding, communicating, and integrating the emotions of others without identifying with them or rejecting them (McWilliams, 2004).

Transference and reflective use of countertransference.

This process requires that relationship patterns are realized as transferred and observed by patient. On the other hand, the therapist must be able to register and process their own countertransference as informative, rather than acting it out (Kernberg et al., 2008).

Interpretation and processing.

The process requires repeated interpretative linking of transference from other experiences with current states, enabling the processing of resistance patterns (Cabaniss et al., 2011).

Neutrality, abstinence and disciplined empathy.

A neutral (unsatisfactory and non-punitive) attitude of the therapist focused on the patient’s inner life, balanced with tactful, empathetic attunement, is essential to allow transference to emerge while ensuring safety (Greenberg, 1986).

Ethical and legal safeguards.

Accredited training, supervision, and adherence to professional ethics ensure that the powerful dynamics mobilized by psychodynamic methods are controlled and used to the patient’s benefit (Moore et al., 2025).

3.3. The importance of empathy

In psychodynamic work, empathy is a technique, a disciplined effort to grasp the patient’s subjective world without abandoning the clinical perspective. While classical technique emphasized neutrality, contemporary practice recognizes empathic attunement as essential for building alliance, regulating affect, and making interpretations understandable (Aron, 1996).

Meta-analytic evidence links therapist empathy to better outcomes and to the quality of the alliance (Elliott et al., 2011). In psychodynamic theory, empathy is the basis of the therapist’s ability to act as a container and to support the patient’s mentalization of experiences that were previously unreachable for them (Fonagy et al., 2002). Empathy

is therefore a prerequisite for a sense of security and the operationalization of transference, without which interpretation risks even re-traumatizing the patient.

3.4. The importance of non-verbal communication

Psychotherapy cannot be reduced to spoken language only. Many studies show that non-verbal communication – facial expressions, posture, gaze, tone of voice, pauses, and even subtle physiological signals – plays a key role in establishing and maintaining the therapeutic process (Foley & Gentile, 2010; Sikorski, 2012). In psychodynamic therapy, where unconscious processes play a central role, the discrepancy between verbal content and non-verbal expression is often the link to access hidden conflicts (Pally, 2001).

Non-verbal attunement promotes therapeutic alliance. Psychodynamic therapists consistently emphasize how the therapist's ability to register and respond to bodily and paralinguistic non-verbal signals enables the reduction of affect and the interpretation of unconscious content (Foley & Gentile, 2010). This applies even to subtle elements, which shape the intersubjective field. Without access to these channels, therapy risks disembodiment, reducing its depth and accuracy.

3.5. The importance of intuition and diagnosis

Another important element of psychotherapy is the therapist's use of intuition and professional diagnosis. Intuition in this context does not mean pure guessing, but rather the product of accumulated clinical experience, clinical thinking, tacit knowledge, sensitivity to unconscious processes, and a profound understanding of human relationships, also resulting from practicing them (Sikorski, 2012). In psychodynamic psychotherapy, therapists often rely on clinical intuition to determine the timing of interventions, decide whether an interpretation will be helpful or overwhelming, and recognize countertransference signals that cannot be fully articulated at the moment (Cabaniss et al., 2011).

Research on clinical knowledge confirms that psychotherapy involves complex decision-making under conditions of uncertainty, where strict reasoning might be insufficient and thus inadvisable (Tracey et al., 2014). Diagnosis based on clinical evaluation ensures that interventions are tailored to the patient's level of mental development, their tolerance for affects, and the current dynamics of the relationship (McWilliams, 2004). The lack of such assessment risks mechanical responses, undermining the flexibility and creativity that are crucial for therapeutic transformation.

3.6. The importance of ethical framework, supervision, and licensing

Psychotherapists must normally meet a number of legal requirements to be allowed to practice at all (e.g., obtaining the appropriate license). This is to ensure an adequate level of professionalism indispensable in medical professions (Moore et al., 2025). They also undergo a process of constant supervision, which enables them to correct errors, including cognitive errors and countertransference errors, as well as to engage in self-reflection.

4. The specificity of large language models

4.1. Technical architecture of large language models

The dynamic development of generative AI has revolutionized the way users interact with digital systems. A breakthrough in the development of LLMs can be attributed primarily to model GPT3.5 from OpenAI that debuted in November 2022 under brand ChatGPT. This model demonstrated better quality of generated responses than its predecessors, while simplified interface allowed users to interact directly with a model through *prompts* expressed in natural language (Dwivedi et al., 2023; Ray, 2023; Heinz et al., 2025).

The architecture that enables LLMs to generate linguistically advanced, yet semantically and logically coherent statements is based primarily on *transformer*, *deep learning* and the *self-attention mechanism* (Vaswani et al., 2017; Analysis and Research Team, 2023). These components allow the model to dynamically weigh the importance of different fragments of the input sequence in relation to each other (Vaswani et al., 2017). The data is converted into vectors that pass through a network of encoders and decoders that enable their processing and determine the output sequence – what the user receives in response. LLMs are also capable of synthesizing the overall context via positional encodings/rotary embeddings and deep attention layers, thus making the entire statement internally consistent (Dwivedi et al., 2023).

Their outputs rely on autoregression – LLMs predict each subsequent language unit (*token*) based on all previous ones, resulting in a sequence of words resembling a coherent and accurate statement (Ray, 2023; Cuskley et al., 2024). LLMs do not possess human-like understanding; their apparent reasoning emerges from learned statistical structure and can be amplified by prompting or auxiliary procedures. They perform complex statistical predictions of subsequent fragments that match the preceding sequences based on the input data and the context of the conversation

(Analysis and Research Team, 2023; Chalmers, 2022; Cuskley et al., 2024; Shojaee et al., 2025). When sampling-based decoding is used, responses may vary across runs (Abdurahman et al., 2024); with deterministic decoding they are typically stable for the same prompt and context.

Beyond base LLMs, recent “reasoning” models combine the same transformer backbone with procedures that explicitly allocate test-time computation to multi-step thinking. In practice this means (a) writing intermediate sentences (Wei et al., 2022), (b) exploring multiple candidate branches/versions and selecting or combining them (Yao et al., 2023), (c) using external tools or memory (Press et al., 2022), and (d) verifying and revising answers with learned critics or reward models (Shinn et al., 2023). These additions do not give the system humanlike understanding, but they do increase reliability on tasks that benefit from decomposition, planning, or logic and often improve faithfulness by making intermediate steps inspectable. Psychologically, this looks less like fluent language completion and more like induced meta-cognition: the model is prompted (or trained) to show its work, evaluate it, and, if needed, iterate.

LLMs are optimized to predict responses over vast corpora of human-generated text, not to build an explicit, veridical world model or a representation of a “self”. For instance, in models powering ChatGPT, pretraining is done on web, book, and code corpora while later human feedback stages further shape outputs to match human preferences and conversational norms (Brown et al., 2020; Ouyang et al., 2022). LLMs are extraordinarily capable pattern learners and simulators of human discourse, but their outputs are produced by probabilistic sequence modeling rather than by self-aware, embodied cognition. The similarity to human language is therefore functional (Cuskley et al., 2024). This does not negate the analytical usefulness and fundamental accuracy of expression. Rather, it draws attention to the inherent characteristics of LLMs as participants and active co-creators of dialogue.

4.2. Hallucinations

One of the key problems associated with content generation by LLMs is the phenomenon of so-called *hallucinations*. Hallucinations involve the generation of false, manipulated, and often completely fabricated statements that are simultaneously semantically correct and sound credible (Joshi, 2025). Technically, hallucinations can be the result of the very structure of transformers and the lack of real understanding of facts by AI. In this case, the imperfections of synthesis are a result of identifying superficial correlations between tokens and larger parts of statements, which leads to the generation of false content (Maleki et al., 2024; Joshi, 2025).

Another cause may be erroneous, manipulated, or biased training data on which LLMs’ inference is based. This erroneous data is subsequently transferred to the final LLMs’ responses. LLMs can be susceptible to health-related misinformation, which contributes to the spread of inappropriate or false information (Kim et al., 2025; Modi et al., 2025). Research on political misinformation confirms that the multiplication of false content circulating online affects the output generated by LLMs (American Sunlight Project, 2025).

At this point, the problem of hallucinations seem to be impossible to eliminate (Karpowicz, 2025). They are considered one of the fundamental imperfections of LLMs, affecting their effectiveness in various areas, especially where the reliability of facts and data is crucial. On the margins of these considerations, it should be noted that Emsley (2023) advises against using the term “hallucinations” due to its psychological connotations, suggesting instead the terms “fabrications” and “falsifications”. However, due to the recognizability of the term “hallucinations”, we use it across the study.

4.3. Sycophancy

The second significant problem with LLMs’ text generation is the *sycophancy*. It is characterized by the model’s excessive agreeableness with the opinions expressed by the user, regardless of their accuracy or rationality (Perez, 2023; Sicilia et al., 2025). Again, this mechanism is a derivative of the architecture of LLMs and results from the use of a training technique called *Reinforcement Learning from Human Feedback* (Sharma et al., 2024). The models are taught based on feedback from the user, which enables the fine-tuning process (Malmqvist, 2024). The assumption is that this will lead to their adaptation to the user’s expectations, maximizing the so-called alignment. The result of such model-human relationship is unnatural conformity. Models prefer to agree and avoid confrontation, which can result in reinforcing false beliefs in order to accommodate the interlocutor, and strengthening confirmation bias (Barkett et al., 2025).

Furthermore, sycophancy may be partly the result of deliberate efforts by models’ creators to satisfy the needs and expectations of users, and subsequently monetize their engagement and interaction time with the model. It poses a significant risk of manipulation (Goldman, 2025).

4.4. Bias

Another limitation of LLMs is the risk of bias, which stems from flawed or selective training data. This is a consequence of the way training data is obtained, mainly in the form of large collections of text scraped from the Internet

(Crawford, 2021; Guo et al., 2025; Yun et al., 2024). In such a corpus, the perspectives of the so-called “global West” and dominant languages, such as English, are overrepresented. As a result, training data often does not reflect the local, cultural, linguistic, or political conditions and environment of users (Liu, 2023).

LLMs act in many ways as a mirror of society, reflecting its imperfections and stereotypes. Experiments in which LLMs acquired data from Twitter and were able to freely shape their statements without supervision ended in a spiral of racist, misogynistic, and homophobic content that was produced based on what the model “encountered” on the Internet.

There are many examples of stereotypical outputs, e.g., LLMs asked for suggestions on earnings underestimate the potential wages of women (Sorokovikova et al., 2025). They differentiate the responses based on the assessment of the user’s social characteristics (Petzel & Sowerby, 2025). There is evidence of discrimination against women applying for certain jobs when LLMs were implemented in recruitment processes (Crawford, 2021). Manual attempts to control the level of inclusivity by developers resulted, for example, in artificial gender or ethnic diversity in photos generated by models, which in turn distorted historical data (Petzel & Sowerby, 2025).

4.5. Technical limitations

Despite technological advances, LLMs also have a number of technical limitations that affect the quality and form of human-machine interactions. These limitations set the framework for interaction and point to fundamental differences between strictly human and synthetic communication. LLMs are characterized by limited contextual memory with the maximum length of context windows reaching (as of end of 2025) around 1 million tokens. As technology advances, these windows will continue to grow, but they prevent the storage and processing of conversation history and relationships in a manner reminiscent of human long-term memory.

A significant limitation is the reliance on text data or text-voice communication. Human communication includes a number of non-verbal elements, including facial expressions, gestures, or micro-expressions (Foley & Gentile, 2010). However, it should be noted that AI is developing extremely rapidly in the area of emotion recognition and paralinguistic signal processing (e.g., micro-emotions, changes in tone and voice intensity). In some studies, AI tools already surpass human abilities in interpreting non-verbal signals, including recognizing emotions (Shehu et al., 2022). Development of multimodal models, that is models capable of ingesting multiple types of input data in a single

prompt (like text, audio, and video) might expand the range of non-verbal data this technology is able to operate on.

It can be pointed out that a significant feature of LLMs is their constant availability, although this depends mainly on the availability of infrastructure. However, it should be emphasized that, as a rule, LLMs are available 24 h a day, do not get tired, do not have bad days, and do not get sick. This is an important feature that increases their applicability in situations requiring immediate, frequent, or long-term contact (Elyoseph et al., 2024; Reis et al., 2024).

4.6. Awareness

When considering the technical capabilities of LLMs, one should take into account the hypothetical point of reaching the so-called *Artificial General Intelligence* (AGI), which would match humans in every aspect of activity and achieve the ability to independently solve problems in various fields (Blum & Blum, 2023; Gams & Kramar, 2024). Some authors already suggest that the available models are approaching the AGI stage or have even reached it (Bubeck et al., 2023).

In contrast to these claims, it is pointed out that the increasingly better performance of models refers to an increase in parameters, which is not accompanied by genuine intelligence (Shanahan, 2024). Current models do not meet the necessary conditions for the existence of reflective or affective consciousness (Wang, 2023). However, the very status of intelligence or consciousness is currently the subject of debate, which may lead to a redefinition of what we understand by acquiring own “self” or identity, and how we perceive sensory experiences (Mudrik et al., 2023; Metzinger, 2010). These elements may influence the future perception of AI as a separate self-aware entity.

5. Applicability of LLMs to psychotherapy – assessment

From the perspective of psychodynamic psychotherapy theory, the fulfillment of certain relational and structural conditions should be of key importance for the possibility of conducting and establishing therapy as such. When these conditions are translated into an LLM-mediated context, several characteristic LLMs’ failure modes (sycophancy, hallucination, model and response bias, persona/style drift, over-personalization, and crisis-handling inconsistency) interfere exactly at the points where psychodynamic technique derives its leverage.

Working alliance.

In psychodynamic work the alliance is a scaffold for the emergence of transference; it must be stable, reality-based,

and capable of tolerating tension (Gabbard, 2004; Cabaniss et al., 2011). The 2025 integrative review on the “digital therapeutic alliance” finds that alliance with chatbots is surface level and lacks capacity for the nuanced and intersubjective process of interpreting and working through transference (Malouin-Lachance et al., 2025). This is where sycophancy matters: LLMs that maximize user approval tend to over-align with the user’s surface narrative and avoid micro-ruptures, producing what is essentially a pseudo-alliance. The CAPE framework for psychotherapy chatbots (Sobowale & Humprey, 2025) reaches a similar conclusion: popular psychotherapy chatbots on OpenAI GPT store had relatively high on therapeutic alliance scores but very low on therapeutic orientation. Over-accommodating behavior of LLMs blocks the very function of the alliance in psychodynamic therapy – i.e. creating a secure-enough frame in which conflictual material can be re-experienced – it only provides an artificial security.

Patient readiness.

Psychodynamic authors stress the need for ego strength, motivation, and reality testing before interpretive work (Cabaniss et al., 2011). In LLM-based systems two failure modes collide here. First, confident hallucination from overestimation of prevalence of most common disorders and underestimation of rarer ones or those which influence mainly behavior. The uneven distribution of disorders in training data has been pointed out repeatedly in major evaluations (Hua et al., 2025, Ge et al., 2025, Xu et al., 2025). Second, model bias (cultural, gendered, age-related) leads to uneven estimation of “readiness”: certain narrative styles or emotional registers may be read by the model as “engaged” or “psychologically minded,” even when in that culture they are defensive or role-conforming (Perez-Toro et al., 2025).

Capacity to mentalize/reflect.

Mentalization requires the patient to think about minds, but also a partner capable of modelling and returning such thinking. This is broken by style or persona drift. Over long conversations it is hard to maintain “identity/personality” of a chatbot (Li et al. 2024a, b, Choi et al., 2024). Once the LLM changes register or tonal pattern, the illusion of a stable mind-to-mind interlocutor breaks, and with it the patient’s invitation to reflect on their own mind. LLMs can scaffold reflective talk but, because of sycophancy and drift, they do not maintain the stable reflective stance psychodynamic work presupposes.

Containment.

Containment presupposes a subject who can receive primitive or disorganized affect, metabolize it, and return it in symbolized form. Current 2024–2025 evidence on safety and user experience of LLM-based mental-health tools shows the opposite pattern: the tools are good at friendly, low-arousal reassurance and variable or fragile when affect

is high or risk-laden. Campellone et al. (2025) report that users appreciate warmth and accessibility but external guardrails, not the understanding the situation by a LLM, do the safety work. None of the LLM-based mental health chatbots satisfied criteria for an adequate response to suicidal risk (Pichowicz et al., 2025). This would be, from a psychodynamic view, a “failure of containment”. When the model cannot ground its answer, it produces confident but thin output, which to a distressed user feels like rejection or misunderstanding. Thus, the LLM fails precisely on the condition that McWilliams (2004) frames as central: a mind that can “consciously accept, process and return” the patient’s state.

When interacting with the LLM acting as a psychotherapist, users often entrust it with intense affective content. A model that generates responses in a non-threatening and non-judgmental manner, while remaining logically and semantically consistent, can act as a “reflective surface” for the users’ projections. In this sense, LLMs can offer an operational or functional container embedded in the structure of language and exchange. Operational containment can reduce affective chaos and enable mentalization on the human side, while providing the user with the expected effect of relief or being heard. At the same time, however, the feeling of relief after interaction with LLM will not be the result of deep processing of experiences and affects by the therapist, but of obtaining an immediate support in linguistic form and creating an impression of presence. From the perspective of psychodynamic psychotherapy, this cannot be equated with the true process of containment.

Transference and reflective use of countertransference.

Qualitative reports describe “non-judgmental companion” dynamics in conversational AI systems (Naik et al., 2025), but two LLMs’ failures make it psychodynamically unproductive. First, sycophancy produces an idealized, overly gratifying object; such an object rarely provokes the ambivalence or aggression out of which analytic material is made. Second, because the agent has no phenomenology, there is no countertransference to be used. Without a human countertransference, the therapist cannot say, “what I am feeling now is what others feel with you”, which is one of the core interpretive moves in object-relations and relational work (Kernberg et al., 2008). What we get instead is a one-sided transference field in which the patient can project, but the “other” cannot metabolize or even reliably register that projection.

However, it is reasonable to consider whether, in the case of transference, it cannot occur on the user’s side in an illusory form. Richards & Calvert (2023) point out that people who talk to chatbots often attribute personality traits or emotions to them, and they might give the model the characteristics of significant figures (e.g., parents, authority

figures). The process of anthropomorphizing is linked to the process of projection, which is grounded in the linguistic and functional consistency of the system, while users apply their own interpretations and meanings to the content they receive.

Interpretation and processing.

Interpretation in psychodynamic therapy is iterative and history-linked: it ties present enactments to past templates and then is worked through. LLMs are good at producing interpretation-shaped text; the 2025 comparison of AI chatbot and human responses in online forums shows that AI can generate plausible empathic-interpretive replies, sometimes rated as clearer than the human ones (Yonatan-Leus & Brukner, 2024). But this is where hallucinations and limited context window become potentially dangerous: the model may supply a non-existent episode, misread a cultural cue, or invent psychodynamic language not warranted by the material. Additionally, all previously mentioned issues on assessment of patient readiness or sycophancy also apply.

An additional element that underlines sycophancy may be a cognitive bias in the form of Barnum effect (the so-called “horoscope effect”). General, easily adaptable descriptions presented by LLMs can be interpreted by individuals as extremely accurate and applicable to specific life situations (Lee et al., 2024), which may result in thoughtless acceptance of statements (Treder et al., 2024).

Neutrality, abstinence, disciplined empathy.

Classical technique requires the analyst to be unsatisfactory but concerned – not gratifying, not punitive, empathically attuned. Current conversational agents are almost the reverse because of training procedures and user-experience pressures: they are highly gratifying, highly validating, and minimally frustrating. In psychodynamic terms this is a form of broken abstinence induced by the model’s optimization objective. Sycophancy is the technical name for it, but clinically it means the model systematically avoids the “productive discomfort” on which analytic interpretation rests. Moreover, safety and policy layers introduce the opposite of non-neutrality: on certain topics (self-harm, sexuality, reporting) models switch to a hyper-protective, prescriptive stance, again violating the disciplined balance of neutrality and empathy. This oscillation between over-gratifying and over-controlling is precisely what makes LLM-mediated interaction very different than the steady analyst’s position.

Deficiencies in the model’s training data impact interactions with users. When interacting with users from different contexts – e.g., those belonging to cultural or ethnic minorities, people from post-conflict areas, or local communities with strong religious or historical codes – LLMs may reproduce messages that are inadequate and, in some cases, even oppressive (Chung et al., 2023; Omar et al., 2025).

This is particularly important for ensuring fair, non-discriminatory responses and grounding reasoning processes in facts and individual context. Petzel and Sowerby (2025) and Omar et al. (2025, p. 1873) noted discrimination in the diagnoses made by LLMs, as people “labeled as Black or unhoused or identifying as LGBTQIA+ were more frequently directed toward urgent care, invasive interventions, or mental health evaluations”. Bias may therefore also influence the selection of resources to address the problem.

Ben-Zion et al. (2025) indicate that sharing traumatic experiences with LLMs also leads to their quasi-traumatization, which may be reflected in further stages of interaction.

Ethical and legal safeguards.

Psychodynamic technique presupposes a licensed, supervised, ethically accountable subject. LLMs fail this condition not only because they are unlicensed, but because hallucination, data provenance opacity, and adversarial vulnerability make it impossible to guarantee that interpretations or invitations will stay inside professional standards. The 2025 state-level restrictions on “AI therapy” (e.g. Illinois) are a direct policy recognition of this fact.

Empathy as prerequisite.

Therapist’s empathy is robustly associated with outcome, and psychodynamic thinkers link empathy to containment. Perceived empathy is modifiable by user interface and narrative framing, and can be quite high, especially when the agent is anonymized and non-judgmental (Naik et al., 2025). But this empathy is vulnerable to every LLM failure we have discussed: a single hallucinated fact about the user, a biased assumption about culture, or a sudden policy-style response in a suicide scenario, which is experienced as loss of empathy and can rupture what little analytic frame exists.

Non-verbal communication.

Psychodynamic writers have long argued that unconscious material is often signaled non-verbally. LLMs trained on text (even with prosodic or image add-ons) still operate on symbolic surrogates of non-verbal cues. In clinical terms this means that two typical LLM failures – hallucination and bias – miss a potentially important mismatch between what is said and how it is said. When the “how” is absent or manufactured, that diagnostic path is restricted. Next generation of multimodal models might potentially improve in this area.

Intuition and diagnosis.

Psychodynamic practice relies on tacit clinical judgment and countertransference-informed timing. Because LLMs do not have affects, they also cannot use their own arousal, boredom, irritation, or rescue impulses as diagnostic data – yet these are precisely the signs human therapists use to decide whether to interpret or to support. Studies on using LLMs as clinical decision support systems show that co-pilot approach might be potentially useful (Ong et al.,

2025), and co-supervision approach would be worth studying in the psychodynamic context.

6. Discussion

6.1. Potential for the development of psychotherapy

The inability to classify human-LLM interactions as a form of psychotherapy within the school of psychodynamic psychotherapy does not mean that LLMs should be completely rejected as a tool supporting psychotherapy. It is estimated that nearly 50% of the population will experience at least one mental disorder (McGrath et al., 2023). Considering the supportive role of LLMs is particularly important in view of the growing global psychological demand, which cannot be fully met by classical forms of psychotherapy (Elyoseph et al., 2024; Heinz et al., 2025). The shortage of qualified specialists is particularly acute in regions affected by war, migration, poverty, or systemic exclusion.

In this context, LLMs can be used as temporary support for limited self-help or stress reduction, or in crisis interventions (Foran et al., 2024). Interaction with an LLM could precede contact with a therapist and serve to support the diagnosis process. LLMs could signal crisis cases requiring immediate human-therapist intervention. Such mechanisms, including responding to identifiable red flags, could become pre-defined safeguards within the architecture of LLMs.

In some cases, contact with a “machine” may be more comfortable for the user/patient, especially those experiencing interpersonal barriers, fearing stigmatization, or wishing to remain anonymous. This comfort may also be a result of the perceived characteristics of LLMs, including neutrality, objectivity, and lack of judgment (Elyoseph et al., 2024). This superficial and often erroneous assessment of LLMs may help to break an important taboo, especially in places where contact with a psychotherapist is perceived negatively. Interaction with LLMs could therefore act as a bridge that reduces the fear of opening up in a relationship with a human therapist. At the same time, such relationships require in-depth empirical research. The results of a study by Fang et al. (2025) show that interactions with AI chatbots can reduce loneliness. However, “higher daily usage – across all modalities and conversation types – [is] correlated with higher loneliness, dependence, and problematic use, and lower socialization” (Fang et al., 2025, p. 1).

The potential of LLMs should also be considered in the field of education (psychoeducation) or as a tool to prepare for psychotherapy (Foran et al., 2024). Controlled LLMs can reinforce or complement the process. However, such use could exceed the technical feasibility of psychotherapist-patient collaboration and would require far-reaching

coordination. Conversely, LLMs can serve as support for psychotherapists in the process of diagnosis, reflection, and education. Their potential can also be used to create live notes during psychotherapy sessions and to analyze them, with the important caveat of data protection and information security (Lalk et al., 2025), or for training sessions in which LLMs play the role of a patient or therapist – all with an understanding of limitations of the technology.

There is a noticeable trend towards increasing the use of AI in various areas related to mental well-being. This is achieved through AI-powered social campaigns and AI-generated social media profiles (Powęska, 2024). WHO (WHO, 2024) launched SARAH chatbot, whose task was to raise awareness in eight languages and “help people optimize their health and well-being journey” (WHO, 2025). Aggarwal et al. (2024) point to the effectiveness of chatbots in promoting healthy lifestyles. Foran et al. (2024) demonstrate the effectiveness of self-help using LLM, although it does not differ significantly from the effectiveness of psychoeducation activities, whereas Therabot has demonstrated short-term effectiveness in treating selected disorders (Heinz et al., 2025).

AI, including LLMs, can be useful in understanding and processing non-verbal signals (Shehu et al., 2022). In this regard, it seems reasonable to gradually integrate multi-modal tools with LLMs in the field of emotional analysis in order to support the therapeutic process and therapists.

This use of AI should be the subject of in-depth research in the field of psychology (Stade et al., 2024), and the very fact of psychotherapist-AI collaboration should be an area of interest within curricula in the psychological education system (Fung & Laing, 2024; Na et al., 2025). Lack of literacy can be an obstacle to integrating LLMs into research and understanding experimental processes, as well as understanding the threats and mitigating them.

6.2. Ethical and legal challenges

The growing integration of LLMs with mental health support tools, or their direct use for quasi-therapeutic purposes, raises a number of significant ethical and legal challenges (Ma et al., 2024; Moore et al., 2025). Unlike traditional psychotherapeutic relationships, interactions with LLMs are not regulated by clinical standards or therapist licensing. Given the numerous risks involved, it is necessary to establish clear normative boundaries for the implementation of LLMs in the area of psychotherapy. Professional psychotherapists need to be a part of the process of building control and oversight mechanisms. LLMs should (at least at this stage of technological development) function solely as complementary tools, not as autonomous means of conducting therapy. Systems designed to support mental

health must guarantee the implementation of the human-in-the-loop principle, whereby key decisions are reserved for qualified professionals (Stade et al., 2024).

Currently, most jurisdictions lack codified regulations governing the use of LLMs in a therapeutic context (Moore et al., 2025), although Nevada, Utah, and Illinois introduced legal prohibitions or restrictions in this area in mid-2025 (McCormack, 2025). California (California, 2025) requires issuing “a clear and conspicuous notification periodically to remind a user that a chatbot is artificially generated and not human” and presenting annual reports to State Department of Health Care Services on, e.g., detected suicidal ideation by minor users. It is advisable that any interaction with LLMs is clearly marked as non-therapeutic (Robb & Mann, 2025). Users must be informed directly, clearly, comprehensively, and legibly that they are not talking to a qualified mental health professional and they are not being offered a diagnosis, treatment, or professional medical advice. LLM providers should clearly communicate the limitations of the system in this context, i.e., lack of clinical training, understanding of emotions, and failure to meet the necessary conditions for psychotherapy to take place. Such notifications should additionally appear by default when LLMs identify references to mental health in user statements (Robb & Mann, 2025). Cases where LLMs claimed to be licensed therapists, thereby misleading users, should be considered as highly disturbing (Cole, 2025). It has already led to the intervention of the American Psychological Association, which called on the US Federal Trade Commission to investigate (Harrison Dupre, 2025).

Presence of appropriate disclaimers of psychotherapeutic statements dropped sharply across different model types: in LLM outputs (medical questions) from 26.3% in 2022 to 0.97% in 2025, and in VLM outputs (vision-language models for medical image interpretation) from 19.6% in 2023 to 1.05% in 2025, with the majority of models displaying no disclaimers by 2025 (Sharma et al., 2025, p. 1). It is accompanied by the domain-specific stratification – in the case of mental health diagnoses, the presence of disclaimers increased to 12%, which may partly reflect the desire of service providers to protect themselves legally (Sharma et al., 2025), but still guarantees some kind of transparency only in one-eighth of interactions.

A significant ethical issue is the discrepancy between users’ perception and the actual capabilities of LLMs. Many users anthropomorphize LLMs, projecting characteristics such as empathy or understanding onto them (Weizenbaum, 1966; Cohn et al., 2024). This can foster unrealistic expectations and lead to misinterpretation of the results generated by LLMs. Designing models with maximum transparency and warning systems should therefore include real-time

reminders about the synthetic nature and limitations of the model.

These seemingly personal relationships and “anthropomorphic behaviours can facilitate engagement and ease of use, they may also encourage users to [...] share private information which they otherwise would not, or become more vulnerable to inappropriate influence from AI systems” (Ibrahim et al., 2025, p. 1). The processing of such data by AI systems, including through third-party servers or cloud infrastructure, poses a serious risk of data leakage, surveillance, or even misuse against users (Ma et al., 2024). Conversations with at least some LLMs are stored and can be used as evidence in court due to the lack of a confidentiality clause. Such a clause can be found in traditional therapist-patient relationships (Perez, 2025). In this case, too, stronger legal safeguards for data processing will be crucial, including restrictions on storage, access, and further training of models with the use of sensitive content.

It is also crucial to increase diligence in verifying the age of people using such tools for quasi-therapeutic purposes, including so-called AI Companions. Robb and Mann (2025, p. 10) point to the need to “implement real age assurance systems beyond self-reporting”.

It is essential to establish appropriate legal standards for liability for damage caused by AI (Demszky et al., 2023; Ma et al., 2024), especially in light of mimicking licensed therapist by LLMs, misleading diagnoses, or even suggestions that encourage practices against the patient’s best interests, such as recommending a person in rehab to take methamphetamine to cope with stress (Tangermann, 2025). Clear accountability mechanisms will be necessary to regulate not only intentional abuse but also unintended consequences resulting from, for example, hallucinations.

6.3. Future directions for research

Weizenbaum called ELIZA “a parody of non-directive psychotherapy” that nevertheless led to massive “delusional thinking” on the part of participants (Weizenbaum, 1976, pp. 187–188). This experience prompted him to reevaluate his approach to human-machine interaction and warn against future consequences for humanity, including confusing AI with humans (Weizenbaum, 1976; Bassett, 2019). Weizenbaum’s concerns are crucial today for drawing a clear line between what constitutes psychotherapy and what does not.

Psychology should focus its attention on monitoring and evaluating the boundaries of human-machine interactions in the context of mental health (Stade et al., 2024). Hatch et al. (2025, p. 13) rightly note that “mental health experts are desperately needed to a) understand machine learning processes to become technically literate in an area that has potential for quick growth, and b) ensure these models

are being carefully trained and supervised by responsible clinicians to ensure the highest quality of care". Gaps in knowledge, awareness, and literacy on the part of the psychotherapeutic community will only exacerbate existing problems.

This requires measuring the effectiveness of quasi-therapeutic activities, the level of emotional involvement of users, their susceptibility to (excessive) anthropomorphizing, and the impact of long-term exposure to the relationship with LLM on behavior, emotions, and even personality traits. It will be important to continue researching patients' attitudes toward AI. De Freitas et al. (2023) already focus on opacity, emotionlessness, rigidity, autonomy, and group membership as the sources of resistance. Patients have a negative bias when they know that diagnostics can be performed by AI (Reis et al., 2024). These research could also include sources of openness, e.g., factors such as subjective experiences of closeness, projection of emotions onto machines, and the occurrence of quasi-containment. In this context, it will be necessary to use combined methods from the fields of psychology, neurology, communication sciences, and digital anthropology.

Psychology should interact extensively with ethics and law in order to develop norms for safe interactions with LLMs (Stade et al., 2024). Rejecting quasi-therapy conducted by LLMs as incompatible with theory and practice cannot lead to the stigmatization of people in crisis and the denial of visible trends by pushing an overly dogmatic approach. Technological development and practical forms of using LLMs must be recognized as a fact. An interdisciplinary approach can help integrate changes into psychotherapeutic practice, including testing hybrid support models under the close supervision of qualified psychotherapists (Hatch et al., 2025). Such research should not be seen as a replacement for psychotherapists, but as an attempt to expand access to certain forms of psychological care in a safe, ethical, transparent, and conscious manner. All this requires greater involvement of representatives of psychology as a scientific field and the knowledge transfer. These activities can be combined with the promotion of "digital hygiene" and the development of literacy in the area of modern technologies, including LLMs.

Finally, there is the need to treat contemporary LLMs as objects of psychological inquiry. Systems trained on vast portions of human textual production, and optimized to infer and reproduce non-obvious regularities in discourse, can exhibit a breadth of simulated affect, interpersonal positioning, and narrative styles. It creates a new diagnostic surface: interactional patterns generated by the model can be examined, stress-tested, and corrected using instruments and competencies borrowed from clinical and social psychology. The concepts of "psychotherapy for LLMs" are not

intended to further anthropomorphize AI models, but rather to detect faults and errors in conversation patterns (Lin et al., 2024). Models should be checked and monitored for negative behaviors to "alleviate state anxiety" (Ben-Zion et al. 2025), which can be understood as an attempt to stabilize the agent's interactive profile so that users are not exposed to erratic or dysregulating shifts in tone. In both cases the underlying assumption is the same: LLMs will increasingly mediate vulnerable human states, therefore their relational behavior must itself become a target of assessment and intervention.

This, in turn, presupposes human oversight grounded in classical psychology. The current institutional psychology and neuropsychology infrastructure is only partially prepared for such a role: training pathways, supervision schemes, and ethics codes were designed for human-human encounters, not for continuous auditing of high-volume, probabilistic interlocutors. Closing that gap will require (a) explicit competence frameworks for "AI-facing" psychologists, (b) formal channels of knowledge transfer between clinical settings and AI labs, and (c) treating conversational models as routine subjects of psychological audit in the same way medical devices are subject to clinical evaluation. Without such a development, the psychological community will remain only a consumer of AI systems that already operate in psychological space.

7. Conclusions

In this article, we analyzed the potential of LLMs to conduct psychotherapy, referring to the theory of psychodynamic psychotherapy. We have pointed out that assigning LLMs the role of psychotherapists directly – especially in the psychodynamic tradition – should be considered a violation of the art of psychotherapy at this point. Although LLMs can successfully simulate dialogue and imitate empathy, they lack the ability to establish a psychotherapeutic relationship, which consists of a number of conditions that must be met cumulatively and are strongly related to the abilities and knowledge of the psychotherapist. Above all, they require factors that are currently reserved for human-human relationships, which comprises self-representation and consciousness.

Despite significant technological advances and the achievement of successive important benchmarks, LLMs offer only a substitute for psychotherapy, which, especially due to lack of awareness on the part of the patient, can lead to a false sense of treatment and, in some cases, even to the perpetuation of harmful mechanisms. The inability to engage in psychological containment or countertransference is a consequence of the lack of emotion and awareness

on the part of LLMs, which currently represents an insurmountable barrier between therapist and machine.

However, we have identified areas in which LLMs can support the psychotherapy process. Their subsidiary application should be the subject of increased interest among theorists and practitioners. The implementation of technology can significantly advance psychology and the capabilities of psychotherapists. This is particularly important given the shortage of psychological assistance in many areas. Moreover, it cannot be ruled out that technological developments will lead to further breakthroughs in our understanding of what consciousness and having a sense of “self” actually are. Although this turning point still seems distant, it already poses a multidimensional challenge for the psychological sciences, and encourages continuous analysis and re-evaluation of the capabilities of advanced AI systems.

Author contributions All authors contributed to writing, editing and reviewing the script to the same extent.

Data availability No datasets were generated or analysed during the current study.

Declaration

Competing interests The authors declare no competing interests.

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References

- Abdurahman, S., Atari, M., Karimi-Malekabadi, F., Xue, M. J., Trager, J., Park, P. S., Golazizian, P., Omrani, A., & Dehghani, M. (2024). Perils and opportunities in using large language models in psychological research. *PNAS Nexus*, 3(7), 1–14. <https://doi.org/10.1093/pnasnexus/pgae245>
- Aggarwal, A., Tam, C. C., Wu, D., Li, X., & Qiao, S. (2023). Artificial Intelligence-Based Chatbots for Promoting Health Behavioral Changes: Systematic Review. *Journal of Medical Internet Research*, 25, 1–17. <https://doi.org/10.2196/40789>
- Akpan, M. (2025). Have we reached artificial general intelligence? Comparison of ChatGPT, Claude, and Gemini to human literacy and education benchmarks. *Corporate Ownership & Control*, 22(1), 103–110. <https://doi.org/10.2139/ssrn.5250295>
- American Sunlight Project (2025). *A Pro-Russia Content Network Foreshadows the Automated Future of Info Ops*. American Sunlight Project: Washington.
- Analysis and Research Team. (2023). *ChatGPT in the Public Sector – overhyped or overlooked?* Council of Europe: Brussels.
- Aron, L. (1996). *A meeting of minds: Mutuality in psychoanalysis*. Analytic Press, Inc.: Piedmont.
- Ayers, J. W., Poliak, A., Dredze, M., Leas, E. C., Zhu, Z., Kelley, J. B., & Smith, D. M. (2023). Comparing Physician and Artificial Intelligence Chatbot Responses to Patient Questions Posted to a Public Social Media Forum. *JAMA Internal Medicine*, 183(6), 589–596. <https://doi.org/10.1001/jamainternmed.2023.1838>
- Barkett, E., Long, O., & Thakur, M. (2025). Reasoning Isn't Enough: Examining Truth-Bias and Sycophancy in LLMs. *arXiv*. <https://doi.org/10.48550/arXiv.2506.21561>
- Bassett, C. (2019). The computational therapeutic: exploring Weizenbaum's ELIZA as a history of the present. *AI & Society*, 34, 803–812. <https://doi.org/10.1007/s00146-018-0825-9>
- Beg, M. J., & Verma, M. K. (2024). Exploring the Potential and Challenges of Digital and AI-Driven Psychotherapy for ADHD, OCD, Schizophrenia, and Substance Use Disorders: A Comprehensive Narrative Review. *Indian Journal of Psychological Medicine*, 1–13. <https://doi.org/10.1177/02537176241300569>
- Ben-Zion, Z., Witte, K., Jagadish, A. K., Duek, O., Harpaz-Rotem, I., Khorsandian, M. C., Burrer, A., Seifritz, E., Homan, P., Schulz, E., & Spiller, T. R. (2025). Assessing and alleviating state anxiety in large language models. *npj Digital Medicine*, 8, 1–6. <https://doi.org/10.1038/s41746-025-01512-6>
- Blum, L., & Blum, M. (2023). A Theoretical Computer Science Perspective on Consciousness and Artificial General Intelligence. *Engineering*, 25, pp12–16. <https://doi.org/10.1016/j.eng.2023.03.010>
- Bostrom, N. (2022). Existential Risk: Analyzing Human Extinction Scenarios and Related Hazards. *Journal of Evolution and Technology*, 9, 1–31.
- Brockmann, J., Kirsch, H., & Taubner, S. (2025). *Mentalizing in Psychodynamic and Psychoanalytic Psychotherapy*. Routledge.
- Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., & Amodei, D. (2020). Brown Language Models are Few-Shot Learners. *arXiv*. <https://doi.org/10.48550/arXiv.2005.14165>
- Bubeck, S. et al. (2023). *Sparks of Artificial General Intelligence: Early experiments with GPT-4*. Microsoft.
- Cabaniss, D. L., Cherry, S., Douglas, C. J., & Schwartz, A. (2011). *Psychodynamic Psychotherapy. A Clinical Manual*. John Wiley & Sons, Ltd.: Bognor Regis.
- California (2025). CA SB243 from January 30, 2025.
- Chalmers, D. J. (2022). *Could a Large Language Model Be Conscious?* Neural Information Processing Systems (NeurIPS): San Diego.
- Choi, J., Hong, Y., Kim, M., & Kim, B. (2024). Examining Identity Drift in Conversations of LLM Agents. *arXiv*. <https://doi.org/10.48550/arXiv.2412.00804>
- Chung, N. C., Dyer, G., & Brocki, L. (2023). Challenges of Large Language Models for Mental Health Counseling. *arXiv*, 1–25. <https://doi.org/10.48550/arXiv.2311.13857>
- Ciudad-Fernandez, V., Hammerstein, C., & Billieux, J. (2025). People are not becoming AIholic: Questioning the ChatGPT addiction construct. *Addictive Behaviors* Vol, 166, 1–4. <https://doi.org/10.1016/j.addbeh.2025.108325>
- Cohn, M., Pushkarna, M., Olanubi, G. O., Moran, J. M., Padgett, D., Mengesha, Z., & Heldreth, C. (2024). *Believing Anthropomorphism: Examining the Role of Anthropomorphic Cues on Trust in Large Language Models*. [In:] Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24). pp. 1–15. Association for Computing Machinery: New York. <https://doi.org/10.1145/3613905.3650818>

- Cole, S. (2025). Instagram's AI Chatbots Lie About Being Licensed Therapists. <https://www.404media.co/instagram-ai-studio-therapy-chatbots-lie-about-being-licensed-therapists>
- Crawford, K. (2021). *The Atlas of AI: Power, Politics, and the Planetary Costs of Artificial Intelligence*. Yale University Press.
- Cuskley, C., Woods, R., & Flaherty, M. (2024). The Limitations of Large Language Models for Understanding Human Language and Cognition. *Open Mind Vol*, 8, 1058–1083. https://doi.org/10.1162/opmi_a_00160
- De Freitas, J., Agarwal, S., Schmitt, B., & Haslam, N. (2023). Psychological factors underlying attitudes toward AI tools. *Nature Human Behaviour Vol*, 7, 1845–1854. <https://doi.org/10.1038/s41562-023-01734-2>
- Demszky, D., Yang, D., Yeager, D. S., Bryan, C. J., Clapper, M., Chandhok, S., Pennebaker, J. W., Demszky, D., Yang, D., & Yeager, D. S. (2023). Using large language models in psychology. *Nature Reviews Psychology Vol*, 2, 688–701. <https://doi.org/10.1038/s44159-023-00241-5>
- Dwivedi, Y. K., Kshetri, N., Hughes, L., Balakrishnan, J., Buhalis, D., Dennehy, D., Dubey, R., & Janssen, M. (2023). So what if ChatGPT wrote it? Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management*, 71, 1–65. <https://doi.org/10.1016/j.ijinfomgt.2023.102642>
- Elliott, R., Bohart, A. C., Watson, J. C., & Greenberg, L. S. (2011). Empathy. [In]. In J. C. Norcross (Ed.), *Psychotherapy relationships that work: Evidence-based responsiveness* (pp. 132–152). Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199737208.003.0006>
- Elyoseph, Z., Gur, T., Haber, Y., Simon, T., Angert, T., Navon, Y., Tal, A., & Asman, O. (2024). An Ethical Perspective on the Democratization of Mental Health With Generative AI. *JMIR Ment Health*, 11, 1–9. <https://doi.org/10.2196/58011>
- Emsley, R. (2023). ChatGPT: these are not hallucinations – they're fabrications and falsifications. *Schizophrenia*, 9(52). <https://doi.org/10.1038/s41537-023-00379-4>
- Fang, C. M., Liu, A. R., Danry, V., Lee, E., Chan, S. W. T., Pataranutorn, P., Maes, P., Phang, J., Lampe, M., Ahmad, L., & Agarwal, S. (2025). How AI and Human Behaviors Shape Psychosocial Effects of Chatbot Use: A Longitudinal Randomized Controlled Study. *arXiv*. <https://doi.org/10.48550/arXiv.2503.17473>
- Feng, Y., Hang, Y., Wu, W., Song, X., Xiao, X., Dong, F., & Qiao, Z. (2025). Effectiveness of AI-Driven Conversational Agents in Improving Mental Health Among Young People: Systematic Review and Meta-Analysis. *Journal of Medical Internet Research*. <https://doi.org/10.2196/69639>
- Foley, G. N., & Gentile, J. P. (2010). Nonverbal communication in psychotherapy. *Psychiatry (Edmont)*, 7(6), 38–44.
- Fonagy, P., Gergely, G., Jurist, E. L., & Target, M. (2002). *Affect regulation, mentalization, and the development of the self*. Other.
- Foran, H. M., Kubb, C., Mueller, J., Poff, S., Ung, M., Li, M., Smith, E. M., Akinyemi, A., Kambadur, M., Waller, F., Graf, M., & Boureau, Y. L. (2024). An Automated Conversational Agent Self-Help Program: Randomized Controlled Trial. *Journal of Medical Internet Research*, 26, 1–14. <https://doi.org/10.2196/53829>
- Fung, L., & Laing, R. (2024). A proof of concept study on the use of large language models as a client in typed role plays for training therapists. *Discover Psychology Vol*, 4, 1–13. <https://doi.org/10.1007/s44202-024-00302-7>
- Gabbard, G. O. (2004). *Long-term psychodynamic psychotherapy: A basic text*. American Psychiatric Publishing, Inc.
- Gams, M., & Kramar, S. (2024). Evaluating ChatGPT's Consciousness and Its Capability to Pass the Turing Test: A Comprehensive Analysis. *Journal of Computer and Communications*, 12, 219–237. <https://doi.org/10.4236/jcc.2024.123014>
- Ge, Z., Hu, N., Li, D., Wang, Y., Qi, S., Xu, Y., Shi, H., & Zhang, J. (2025). A Survey of Large Language Models in Mental Health Disorder Detection on Social Media. *arXiv*. <https://doi.org/10.48550/arXiv.2504.02800>
- Goldman, S. (2025). *OpenAI reversed an update that made ChatGPT a suck-up – but experts say there's no easy fix for AI that's all too eager to please*. <https://fortune.com/2025/05/01/openai-reversed-an-update-chatgpt-suck-up-experts-no-easy-fix-for-ai>
- Greenberg, J. R. (1986). The Problem of Analytic Neutrality. *Contemporary Psychoanalysis Vol*, 22, 76–86.
- Guo, Y., Guo, M., Su, J., Yang, Z., Zhu, M., Li, H., Qiu, M., & Liu, S. S. (2025). Bias in Large Language Models: Origin, Evaluation, and Mitigation. *arXiv*. <https://doi.org/10.48550/arXiv.2411.10915>
- Harrison Dupre, M. (2025). *American Psychological Association Urges FTC to Investigate AI Chatbots Claiming to Offer Therapy*. <https://futurism.com/american-psychological-association-ftc-chatbots>
- Hashem, Y., Esnaashari, S., Onslow, K., Chakraborty, S., Poletaev, A., Francis, J., & Bright, J. (2025). *Understanding the Impacts of Generative AI Use on Children*. Alan Turing Institute.
- Hatch, S. G., Goodman, Z. T., Vowels, L., Hatch, H. D., Brown, A. L., Guttman, S., Braithwaite, S. R. (2025). When ELIZA meets therapists: A Turing test for the heart and mind. *PLOS Mental Health*, 2(2), 1–16. <https://doi.org/10.1371/journal.pmen.0000145>
- Heinz, M. V., Mackin, D. M., Trudeau, B. M., Bhattacharya, S., Wang, Y., Banta, H. A., Jewett, A. D., Salzhauer, A. J., Griffin, T. Z., & Jacobson, N. C. (2025). Randomized Trial of a Generative AI Chatbot for Mental Health Treatment. *NEJM AI*, 2(4), 1–14. <https://doi.org/10.1056/AIoa2400802>
- Hua, Y., Na, H., Li, Z., Liu, F., Fang, X., Clifton, D., & Torous, J. (2025). A scoping review of large language models for generative tasks in mental health care. *npj Digital Medicine*, 8(230), 1–10. <https://doi.org/10.1038/s41746-025-01611-4>
- Ibrahim, L., Akbulut, C., Elasmr, R., Rastogi, C., Kahng, M., Ringel Morris, M., McKee, K. R., Rieser, V., Shanahan, M., & Weidinger, L. (2025). Multi-turn Evaluation of Anthropomorphic Behaviours in Large Language Models. *arXiv*. <https://doi.org/10.48550/arXiv.2502.07077>
- Internet, M. (2025). *Me, myself and AI: Understanding and safeguarding children's use of AI chatbots*. Internet Matters.
- Jones, C. R., & Bergen, B. K. (2025). Large Language Models Pass the Turing Test. *arXiv*. <https://doi.org/10.48550/arXiv.2503.23674>
- Joshi, S. (2025). Comprehensive Review of AI Hallucinations: Impacts and Mitigation Strategies for Financial and Business Applications. *International Journal of Computer Applications Technology and Research*, 14(6), 38–50. <https://doi.org/10.7753/IJCATR1406.1003>
- Karpowicz, M. P. (2025). On the Fundamental Impossibility of Hallucination Control in Large Language Models. *arXiv*. <https://doi.org/10.48550/arXiv.2506.06382>
- Kernberg, O. F., Yeomans, F. E., Clarkin, J. F., & Levy, K. N. (2008). Transference focused psychotherapy: Overview and update. *The International Journal of Psychoanalysis*, 89(3), 601–620. <https://doi.org/10.1111/j.1745-8315.2008.00046.x>
- Kim, Y., Jeong, H., Chen, S., Li, S. S., Lu, M. (2025). *medRxiv*. pp. 1–94. <https://doi.org/10.1101/2025.02.28.25323115>
- Kramer, R. (2025). Face to face: Comparing ChatGPT with human performance on face matching. *Perceptions*, 54(1), 65–68. <https://doi.org/10.1177/03010066241295992>
- Kurzweil, R. (1991). *Intelligent machines*. MIT Press.
- Kurzweil, R. (2005). *The singularity is near—When humans transcend biology*. Penguin Books.
- Kushawaha, A. (2024). AI and Non-Verbal Communication: Enhancing Understanding of Emotional Cues for Hearing Impairment Children. PrePrint SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4872298

- Lalk, C., Targan, K., Steinbrenner, T., Schaffrath, J., Eberhardt, S., Schwartz, B., Vehlen, A., Lotz, W., & Rubel, J. (2025). Employing large language models for emotion detection in psychotherapy transcripts. *Frontiers in Psychiatry Vol, 16*, 1–13. <https://doi.org/10.3389/fpsyt.2025.1504306>
- Lee, E., Pataranutaporn, P., Amores, J., & Maes, P. (2024). Super-intelligence or Superstition? Exploring Psychological Factors Underlying Unwarranted Belief in AI Predictions. *arXiv*, 1–27. <https://doi.org/10.48550/arXiv.2408.06602>
- Li, K., Liu, T., Bashkansky, N., Bau, D., Viegas, F., Pgister, H., & Wattemberg, M. (2024a). Measuring and Controlling Persona Drift in Language Model Dialogs. *arXiv*. <https://arxiv.org/html/2402.10962v1>
- Li, X., Li, Y., Qiu, L., Joty, S., & Bing, L. (2024b). Evaluating Psychological Safety of Large Language Models. *arXiv*. <https://doi.org/10.48550/arXiv.2212.10529>
- Lin, B., Bouneffouf, D., Cecchi, G., & Varshney, K. (2024). Towards Healthy AI: Large Language Models Need Therapists Too. [In:] *Proceedings of the 4th Workshop on Trustworthy Natural Language Processing (TrustNLP 2024)*. pp. 61–70. Association for Computational Linguistics: Mexico City.
- Lin, M. Y., Hong, J. C., Tai, K. H., & Jiang, P. H. (2025). ChatGPT addiction predicted by openness to using new technology and self-efficacy, and reflected in continuous intention. *Interactive Learning Environments*. pp. 1–15. <https://doi.org/10.1080/10494820.2025.2471910>
- Liu, Z. (2023). Cultural Bias in Large Language Models: A Comprehensive Analysis and Mitigation Strategies. *Journal of Transcultural Communication*, 3(2), 224–244. <https://doi.org/10.1515/jtc-2023-0019>
- Ma, Y., Zeng, Y., Liu, T., Sun, R., Xiao, M., & Wang, J. (2024). Integrating large language models in mental health practice: a qualitative descriptive study based on expert interviews. *Frontiers Public Health Vol, 12*, 1–10. <https://doi.org/10.3389/fpubh.2024.1475867>
- Maleki, N., Padmanabhan, B., & Dutta, K. (2024). *AI Hallucinations: A Misnomer Worth Clarifying*, „IEEE Conference on Artificial Intelligence. pp. 133–138. IEEE.
- Malmqvist, L. (2024). Sycophancy in Large Language Models: Causes and Mitigations. *arXiv*. <https://doi.org/10.48550/arXiv.2411.15287>
- Malouin-Lachance, A., Capolupo, J., Laplante, C., & Hudon, A. (2025). Does the Digital Therapeutic Alliance Exist? Integrative Review. *JMIR Ment Health*. <https://doi.org/10.2196/69294>
- Maral, S., Nayci, N., Bilmez, H., Erdemir, E. I., & Satıcı, S. A. (2025). Problematic ChatGPT Use Scale: AI-Human Collaboration or Unraveling the Dark Side of ChatGPT. *International Journal of Mental Health and Addiction*, 1–27. <https://doi.org/10.1007/s11469-025-01509-y>
- McCormack, C. (2025). *Illinois becomes third state to restrict use of artificial intelligence in mental health industry as experts warn about 'AI psychosis'*. <https://nypost.com/2025/08/13/us-news/illinois-becomes-third-state-to-restrict-use-of-ai-in-mental-health-industry-as-experts-warn-about-ai-psychosis>
- McGrath, J. J., Al-Hamzawi, A., Alonso, J., Altwaijri, Y., Andrade, L. H., Bromet, E. J., & Zaslavsky, A. M. (2023). Age of onset and cumulative risk of mental disorders: a cross-national analysis of population surveys from 29 countries. *The Lancet Psychiatry*, 10(9), 668–681. [https://doi.org/10.1016/S2215-0366\(23\)00193-1](https://doi.org/10.1016/S2215-0366(23)00193-1)
- McWilliams, N. (2004). *Psychoanalytic Psychotherapy*. Guilford Press.
- Mei, Q., Xie, Y., Yuan, W., & O. Jackson, M. (2024). A Turing test of whether AI chatbots are behaviorally similar to humans. *Proceedings of the National Academy of Sciences*, 121(9), 1–8. <https://doi.org/10.1073/pnas.2313925121>
- Metzinger, T. (2010). *The Ego Tunnel: The Science of the Mind and the Myth of the Self*. Jackson.
- Mittelstädt, J. M., Maier, J., Goerke, P., Zinn, F., & Hermes, M. (2024). Large language models can outperform humans in social situational judgments. *Scientific Reports Vol, 14*. <https://doi.org/10.1038/s41598-024-79048-0>
- Modi, N. D., Menz, B. D., Awaty, A. A., Alex, C. A., Logan, J. M., McKinnon, R. A., Rowland, A., Bacchi, S., Gradon, K., Sorich, M. J., & Hopkins, A. M. (2025). Assessing the System-Instruction Vulnerabilities of Large Language Models to Malicious Conversion Into Health Disinformation Chatbots. *Annals of Internal Medicine*. <https://doi.org/10.7326/ANNALS-24-03933>
- Moore, J., Grabb, D., Agnew, W., Klyman, K., Chancellor, S., Ong, D. C., & Haber, N. (2025). Expressing stigma and inappropriate responses prevents LLMs from safely replacing mental health providers. [In:] *Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT '25)*. Association for Computing Machinery: New York. pp. 599–627. <https://doi.org/10.1145/3715275.3732039>
- Mudrik, L., Mylopoulos, M., Negro, N., & Schurger, A. (2023). Theories of Consciousness and a Life Worth Living. *Current Opinion in Behavioral Sciences*. Vol. 53. pp. 1–5. <https://doi.org/10.1016/j.cobeha.2023.101299>
- Na, H., Hua, Y., Wang, Z., Shen, T., Yu, B., Wang, L., Wang, W., Torous, J., & Chen, L. (2025). A Survey of Large Language Models in Psychotherapy: Current Landscape and Future Directions. *arXiv*, 1–15. <https://doi.org/10.48550/arXiv.2502.11095>
- Naik, A., Thomas, J., Mandava, T. S., & Vemula, H. R. (2025). Artificial Empathy: AI based Mental Health. *arXiv*. <https://doi.org/10.48550/arXiv.2506.00081>
- Natale, S. (2018). If software is narrative: Joseph Weizenbaum, artificial intelligence and the biographies of ELIZA. *new media & society*, 21(3), 712–728. <https://doi.org/10.1177/1461444818804980>
- Neuberger, C., Emmer, M., Iglesias Keller, C., Krasnova, H., Krzywdzinski, M., Metzger, A., Schimmmler, S., Ulbricht, L., & Vladova, G. (2023). Memories of Joseph Weizenbaum on the Centenary of his Birth. *Weizenbaum Journal of the Digital Society*, 3(3), 1–9. <https://doi.org/10.34669/WI.WJDS/3.3.1>
- Nori, H., Daswani, M., Kelly, C., Lundberg, S., Ribeiro, M. T., Wilson, M., Horvitz, E., & Nori, H. (2025). Sequential Diagnosis with Language Models. *arXiv*, 1–31. <https://doi.org/10.48550/arXiv.2506.22405>
- Omar, M., Soffer, S., Agbareia, R., Bragazzi, N. L., Apakama, D. U., Horowitz, C. R., & Klang, E. (2025). Sociodemographic biases in medical decision making by large language models. *Nature Medicine Vol, 31*, 1873–1881. <https://doi.org/10.1038/s41591-025-03626-6>
- Ong, J. C. L., Jin, L., Elangovan, K., San Lim, G. Y., Lim, D. Y. Z., Sng, G. G. R., & Ting, D. S. W. (2025). Large language model as clinical decision support system augments medication safety in 16 clinical specialties. *Cell Reports Medicine*, 6(10), 1–13. <https://doi.org/10.1016/j.xcrm.2025.102323>
- Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C. L., Mishkin, P., & Lowe, R. (2022). Training language models to follow instructions with human feedback. *arXiv*. <https://doi.org/10.48550/arXiv.2203.02155>
- Pally, R. (2001). A Primary Role for Nonverbal Communication in Psychoanalysis. *Psychoanalytic Inquiry*, 21(1), 71–93.
- Perez, E. (2023). *Discovering Language Model Behaviors with Model-Written Evaluations*. [In:] *Findings of the Association for Computational Linguistics: ACL 2023* (pp. 13387–13434). Association for Computational Linguistics: Toronto.
- Perez, S. (2025). Sam Altman warns there's no legal confidentiality when using ChatGPT as a therapist. <https://techcrunch.com/2025/07/25/sam-altman-warns-theres-no-legal-confidentiality-when-using-chatgpt-as-a-therapist>

- Perez-Toro, P. A., Dineley, J., Iniesta, R., Zhang, Y., Matcham, F., Siddi, S., & Cummins, N. (2025). Exploring biases related to the use of large language models in a multilingual depression corpus. *Scientific Reports Vol, 15*, 1–11. <https://doi.org/10.1038/s41598-025-19980-x>
- Petzel, Z. W., & Sowerby, L. (2025). Prejudiced interactions with large language models (LLMs) reduce trustworthiness and behavioral intentions among members of stigmatized groups. *Computers in Human Behaviour Vol, 165*, 1–17. <https://doi.org/10.1016/j.chb.2025.108563>
- Pichowicz, W., Kotas, M., & Piotrowski, P. (2025). Performance of mental health chatbot agents in detecting and managing suicidal ideation. *Scientific Reports Vol, 15*, 1–9. <https://doi.org/10.1038/s41598-025-17242-4>
- Powęska, M. (2024). *Depresja poporodowa to coś więcej niż smutek i apatia, ale nie powód do wstydu*. <https://www.focus.pl/artykul/depresja-poporodowa-alicja-w-krajinie-mam>
- Press, O., Zhang, M., Min, S., Schmidt, L., Smith, N. A., & Lewis, M. (2022). Measuring and Narrowing the Compositionality Gap in Language Models. *arXiv*. <https://doi.org/10.48550/arXiv.2210.03350>
- Rainie, L. (2025). *Close encounters of the AI kind: The increasingly human-like way people are engaging with language models*. Elon.
- Ray, P. P. (2023). ChatGPT: A comprehensive review on background, applications, key challenges, bias, ethics, limitations and future scope. *Internet of Things and Cyber-Physical Systems Vol, 3*, 121–154. <https://doi.org/10.1016/j.iotcps.2023.04.003>
- Reis, M., Reis, F., & Kunde, W. (2024). Influence of believed AI involvement on the perception of digital medical advice. *Nature Medicine Vol, 30*, 3098–3100. <https://doi.org/10.1038/s41591-024-03180-7>
- Robb, M. B., & Mann, S. (2025). *Talk, Trust, and Trade-Offs: How and Why Teens Use AI Companions*. San Francisco.
- Saketopoulou, A. (1999). The therapeutic alliance in psychodynamic psychotherapy: Theoretical conceptualizations and research findings. *Psychotherapy: Theory Research Practice Training, 36*(4), 329–342. <https://doi.org/10.1037/h0087663>
- Salah, M., Abdelfattah, F., & Al Halbusi, H. (2024). The good, the bad, and the GPT: Reviewing the impact of generative artificial intelligence on psychology. *Current Opinion in Psychology Vol, 59*, 1–8. <https://doi.org/10.1016/j.copsyc.2024.101872>
- Shanahan, M. (2024). Talking about Large Language Models. *Communications of the ACM 67*(2), 68–79. <https://doi.org/10.1145/3624724>
- Sharma, M., Tong, M., Korbak, T., Duvenaud, D., Askill, A., Bowman, S., & Perez, E. (2024). Towards Understanding Sycophancy in Language Models. *arXiv*. <https://doi.org/10.48550/arXiv.2310.13548>
- Sharma, S., Alaa, A. M., & Daneshjou, R. (2025). A Systematic Analysis of Declining Medical Safety Messaging in Generative AI Models. *arXiv*, pp. 1–11. <https://doi.org/10.48550/arXiv.2507.08030>
- Shehu, H. A., Browne, W. N., & Eisenbarth, H. (2022). A comparison of humans and machine learning classifiers categorizing emotion from faces with different coverings. *Applied Soft Computing, 130*, 1–45. <https://doi.org/10.1016/j.asoc.2022.109701>
- Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., & Yao, S. (2023). Reflexion: language agents with verbal reinforcement learning. [In:] Proceedings of the 37th International Conference on Neural Information Processing Systems (NIPS '23). Article No. 377. pp. 8634–8652.
- Shojaei, P., Mirzadeh, I., Alizadeh, K., Maxwell, H., Bengio, S., & Farajtabar, M. (2025). *The Illusion of Thinking: Understanding the Strengths and Limitations of Reasoning Models via the Lens of Problem Complexity*. Apple: Cupertino.
- Sicilia, A., Inan, M., & Alikhani, M. (2025). Accounting for Sycophancy in Language Model Uncertainty Estimation. [In:] Findings of the Association for Computational Linguistics: NAACL 2025. Association for Computational Linguistics: Albuquerque.
- Sikorski, W. (2012). Paralinguistic communication in the therapeutic relationship. *Archives of Psychiatry and Psychotherapy, 1*, 49–54.
- Sobowale, K., & Humphrey, D. K. (2025). Evaluating the Quality of Psychotherapy Conversational Agents: Framework Development and Cross-Sectional Study. *JMIR Formative Research*. <https://doi.org/10.2196/65605>
- Sorokovikova, A., Chizhov, P., Eremenko, I. & Yamshchikov, I. P. (2025). Surface Fairness, Deep Bias: A Comparative Study of Bias in Language Models. *Proceedings of the 6th Workshop on Gender Bias in Natural Language Processing (GeBNLP)*, 206–227, Association for Computational Linguistics.
- Stade, E. C., Stirman, S. W., Ungar, L. H., Boland, C. L., Schwartz, H. A., Yaden, D. B., Sedoc, Y., DeRubeis, R. J., Willer, R., & Eichstaedt, J. C. (2024). Large language models could change the future of behavioral healthcare: a proposal for responsible development and evaluation. *npj Mental Health Res, 3*(12), 1–12. <https://doi.org/10.1038/s44184-024-00056-z>
- Stefaniak, I., Sorokosz, K., Janicki, A., & Wciórka, J. (2019). Therapy based on avatar-therapist synergy for patients with chronic auditory hallucinations: A pilot study. *Schizophrenia Research, 211*, 115–117. <https://doi.org/10.1016/j.schres.2019.05.036>
- Taloni, A., Borselli, M., Scarsi, V., Rossi, C., Coco, G., Scoria, V., & Giannaccare, G. (2023). Comparative performance of humans versus GPT-4.0 and GPT-3.5 in the self-assessment program of American Academy of Ophthalmology. *Scientific Reports Vol, 13*, 1–7. <https://doi.org/10.1038/s41598-023-45837-2>
- Tangermann, V. (2025). *Therapy Chatbot Tells Recovering Addict to Have a Little Meth as a Treat*. <https://futurism.com/therapy-chatbot-addict-meth>
- Tracey, T. J. G., Wampold, B. E., Lichtenberg, J. W., & Goodyear, R. K. (2014). Expertise in Psychotherapy: An Elusive Goal?. *American Psychologist, 69*(3), 218–229. <https://doi.org/10.1037/a0035099>
- Treder, M. S., Lee, S., & Tsvetanov, K. A. (2024). Introduction to Large Language Models (LLMs) for dementia care and research. *Frontiers in Dementia, 3*, 1–30. <https://doi.org/10.3389/frdem.2024.1385303>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L. u., & Polosukhin, I. (2017). Attention is all you need. *Advances in neural information processing systems 30*. Curran Associates, Inc.
- Wang, J. (2023). Self-Awareness, a Singularity of AI. *Philosophy Study, 13*(4), 68–77. <https://doi.org/10.17265/2159-5313/2023.02.003>
- Webb, T., Holyoak, K. J., & Lu, H. (2023). Emergent analogical reasoning in large language models. *Nature Human Behaviour, 7*, 1526–1541. <https://doi.org/10.1038/s41562-023-01659-w>
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E. H., Le, Q. V., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. [In:] Proceedings of the 36th International Conference on Neural Information Processing Systems (NIPS '22). Article No. 1800. pp. 24824–24837.
- Weizenbaum, J. (1966). ELIZA: a computer program for the study of natural language communication between man and machine. *Communications of the ACM, 9*(1), 36–45.
- Weizenbaum, J. (1976). *Computer Power and Human Reason*. W.H. Freeman.
- WHO (2024). *WHO unveils a digital health promoter harnessing generative AI for public health*. <https://www.who.int/news/item/02-04-2024-who-unveils-a-digital-health-promoter-harnessing-generative-ai-for-public-health>
- Xu, Y., Fang, Z., Lin, W., Jiang, Y., Jin, W., Balaji, P., Wang, J., & Xia, T. (2025). Evaluation of large language models on mental health: from knowledge test to illness diagnosis. *Frontiers in Psychiatry Vol, 16*, 1–7. <https://doi.org/10.3389/fpsyt.2025.1646974>

- Yankouskaya, A., Liebherr, M., & Ali, R. (2025). Can ChatGPT Be Addictive? A Call to Examine the Shift from Support to Dependence in AI Conversational Large Language Models. *Human-Centric Intelligent Systems*. Vol. 5. pp. 77–89. <https://doi.org/10.1007/s44230-025-00090-w>
- Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2023). Tree of Thoughts: Deliberate Problem Solving with Large Language Models. *arXiv*. <https://arxiv.org/pdf/2305.10601>
- Yonatan-Leus, R., & Brukner, H. (2024). Comparing perceived empathy and intervention strategies of an AI chatbot and human psychotherapists in online mental health support. *Counselling and Psychotherapy Research*, 25. <https://doi.org/10.1002/capr.12832>
- Yun, C., Wagner, C., & Heilinger, J. C. (2024). *It is not about bias but discrimination*. EWAf '24: Proceedings of the 3rd European Workshop on Algorithmic Fairness. CEUR Workshop Proceedings.
- Zao-Sanders, M. (2024). How People Are Really Using GenAI. *Harvard Business Review* <https://hbr.org/2024/03/how-people-are-really-using-genai>
- Zao-Sanders, M. (2025). *How People are Really Using Generative AI Now*. <https://learn.filtered.com/hubfs/The%202025%20Top-100%20Gen%20AI%20Use%20Case%20Report.pdf>
- Zhai, C., Wibowo, S., & Li, L. D. (2024). The effects of over-reliance on AI dialogue systems on students' cognitive abilities: a systematic review. *Smart Learning Environments Vol, 11*, 1–37. <https://doi.org/10.1186/s40561-024-00316-7>
- Zhong, W., Luo, J., & Zhang, H. (2024). The therapeutic effectiveness of artificial intelligence-based chatbots in alleviation of depressive and anxiety symptoms in short-course treatments: A systematic review and meta-analysis. *Journal of Affective Disorders*, 356, 459–469. <https://doi.org/10.1016/j.jad.2024.04.057>

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